1. Neural networks

1.1. Network sizes

We give information about the size and other details neural networks we have used in our final evaluation as mentioned in Sec 3.4.

Block Size	Layer sizes	Layers
4X4	(256→ 512→ 1024→ 2048→ 3069)	5
6X6	(256→ 512→ 512→ 1024→ 2048→ 3069)	6
8x8	(256→ 512→ 512→ 1024→ 1024→ 2048→ 3069)	7

Table 1. The number of layers and number of neurons in each layer for **P-NN network** (Sec 3.4). We trained neural networks for different block sizes 4X4, 6X6, and 8X8. Each row in the table corresponds to a particular block size and the corresponding network description.

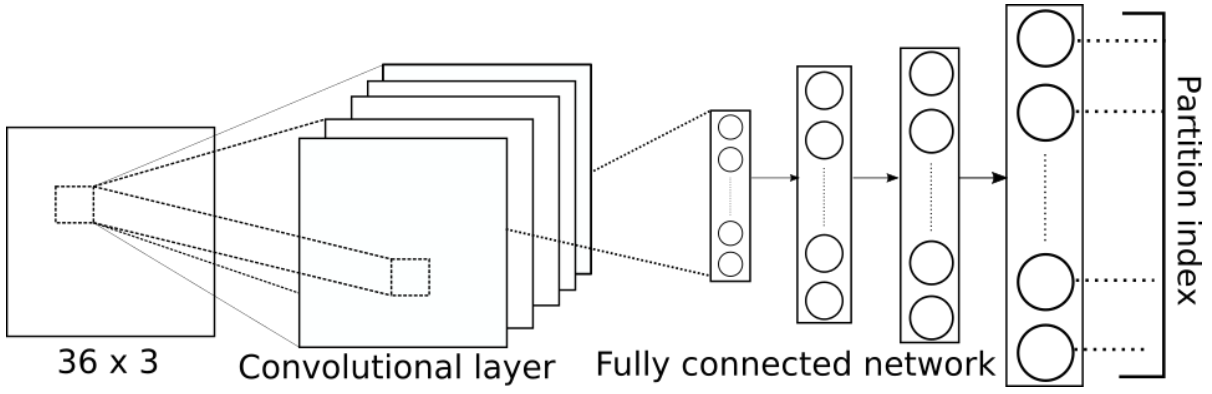
Block Size	Layer sizes	Layers
4X4	(128→ 256→ 512→ 1024→ 1314)	5
6X6	(128→ 256→ 512→ 1024→ 2048→ 3650)	6
8x8	(128→ 256→ 512→ 1024→ 2048→ 4096→ 7154)	7

Table 2. The number of layers and number of neurons in each layer for **DEM-NN network** (Sec 3.4). We trained neural networks for different block sizes 4 X 4, 6 X 6, and 8 X 8. Each row in the table corresponds to a particular block size and the corresponding network description

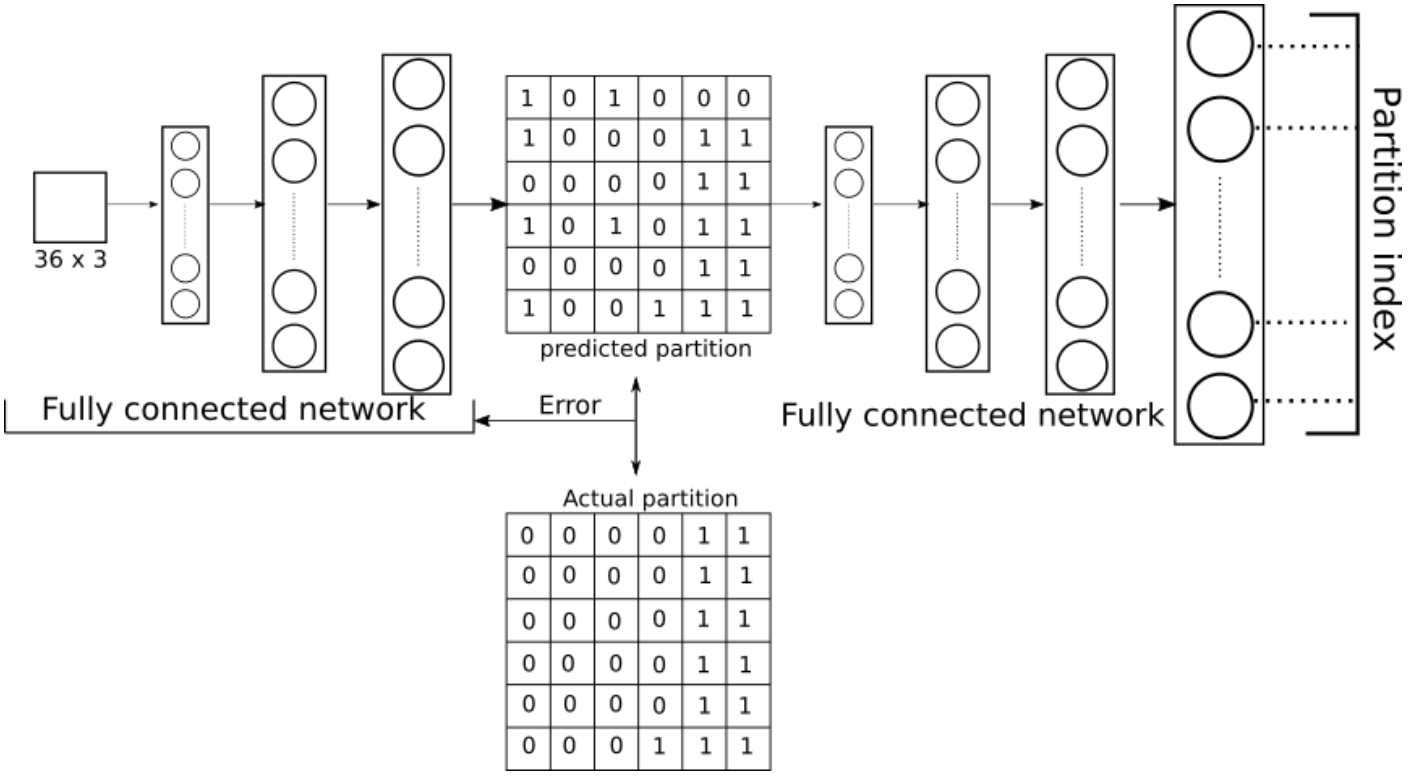
1.2. Network topologies

In addition to the fully connected neural networks we have experimented with other network topologies as well. As mentioned in the paper we have designed a network with additional convolutional layer to training partition index and count. As pointed out in the paper we find that the convolutional neural networks gives prediction as good as original networks.

Convolutional neural networks



Dual loss fully connected network



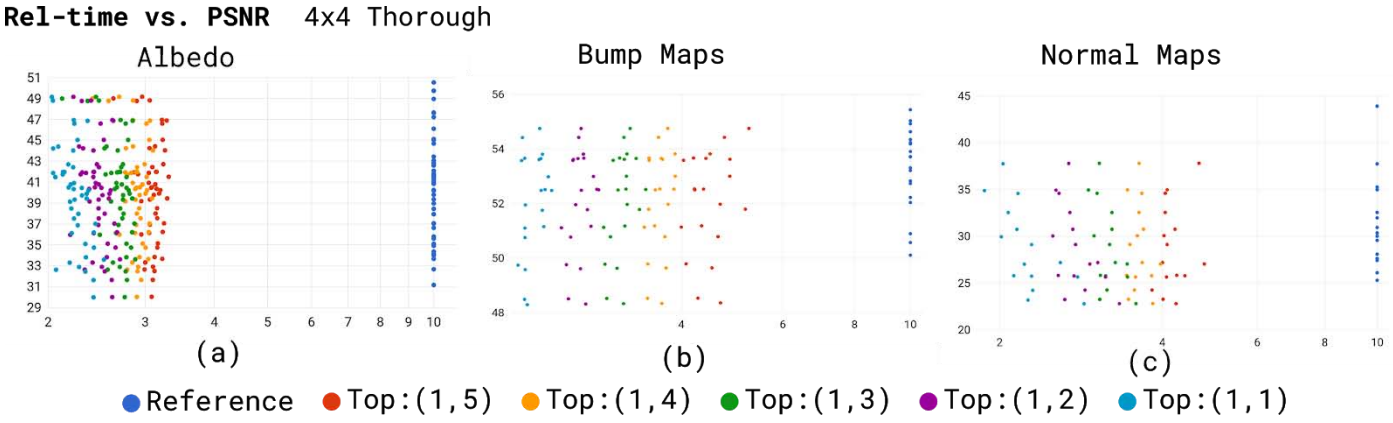
This figure represents the unique dual loss neural network design. In this network we add an additional loss function on the partition structure of the block. We use a hidden layer output to act as a latent variable similar to partition structure. The error from the extra loss function helps the network predict partition indices closer to the actual structure. Although the dual loss network gives better accuracy the network is very large than the P-NN presented in the paper. The gain in accuracy is little compared to the increase in complexity.

2. Evaluation

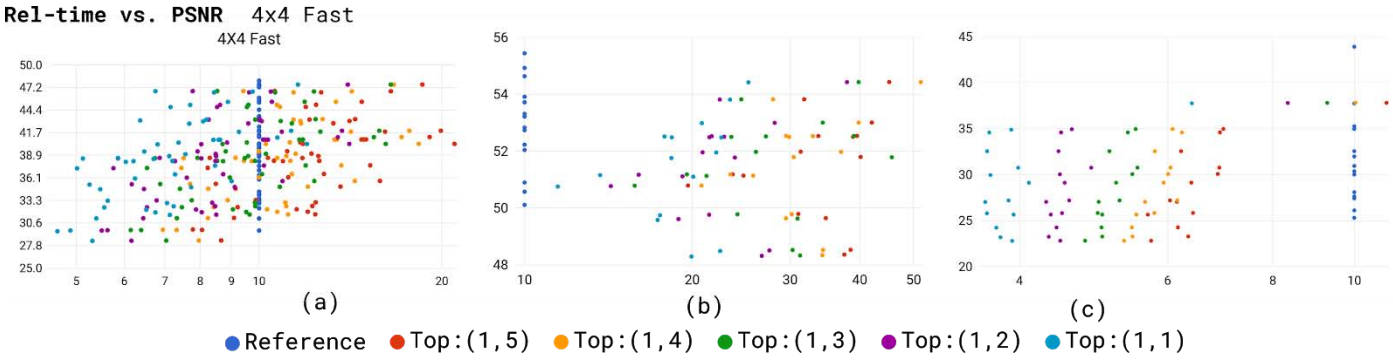
In this section we present additional analysis of the primary results presented in Sec.5 of the paper. We present the encoding-time vs. quality plots for different block-sizes in different modes. We also highlight how the quality changes for a set of images when the encoding-settings are varied.

2.1 Encoding-time vs. quality scatter plots

We present the scatter plots of encoding time vs quality for different encoding settings similar to the plot in the paper (Fig. 7). We notice that at the ref-encoding setting goes from *Thorough* to *Fast* the speed up gained using TexNN is similar to that of ref-encoder. We present the scatter for block sizes 6x6 and 8x8 in Exhaustive, Thorough, and Fast modes. We notice that using our method TexNN we improve the encoding time by a factor of 7-8X with little drop in PSNR

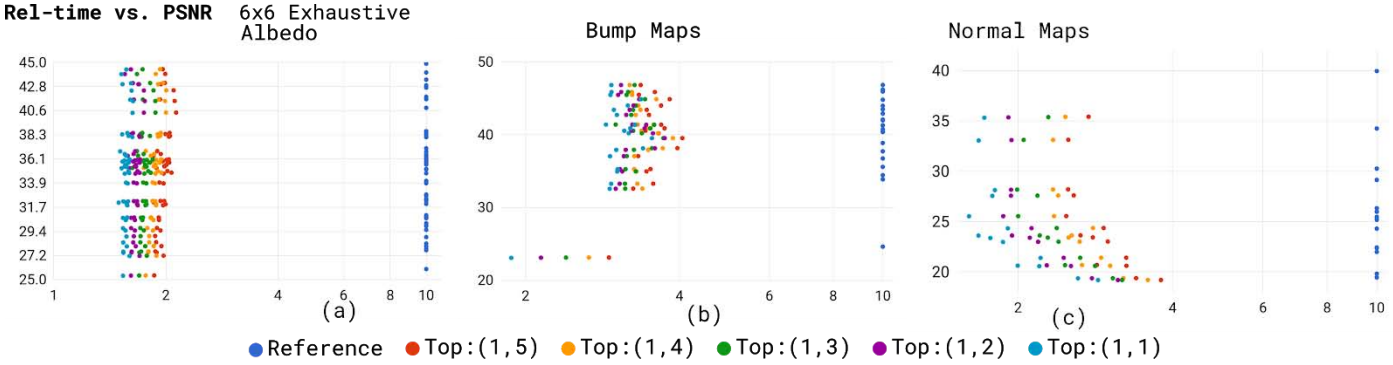


The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in exhaustive setting for block size 4X4. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.

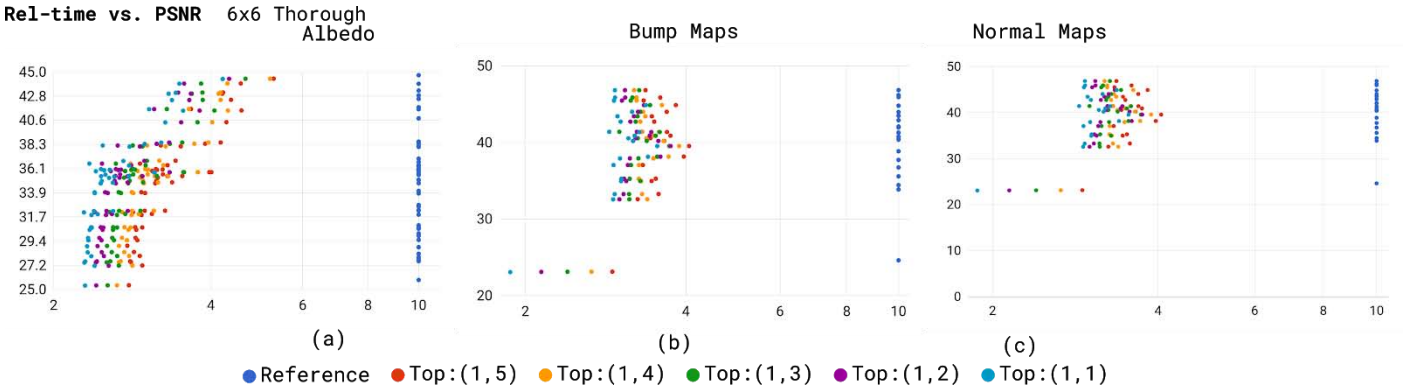


The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in fast setting for block size 4X4. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). In Fast mode the ref-encoder starts to perform better. The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.

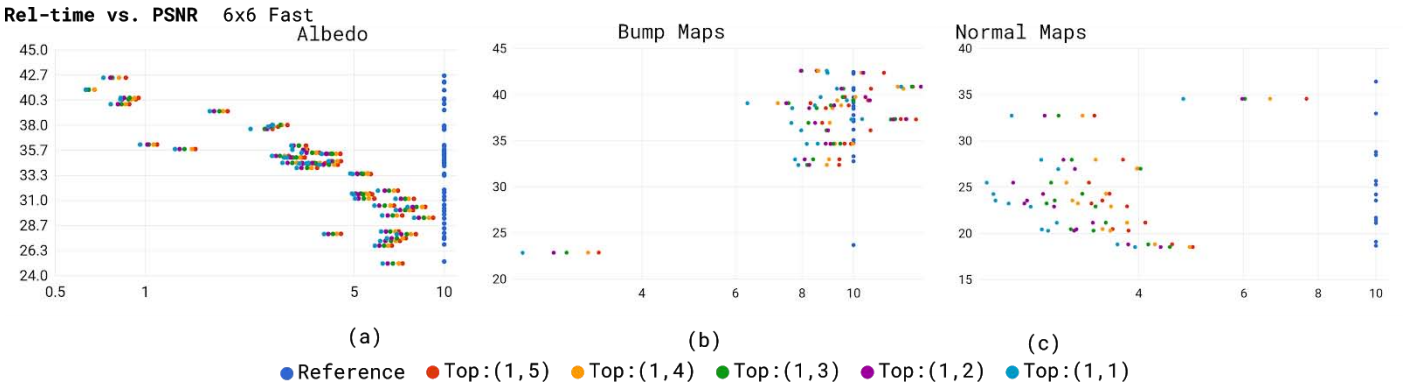
Block-size: 6X6



The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in fast setting for block size 6x6. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.

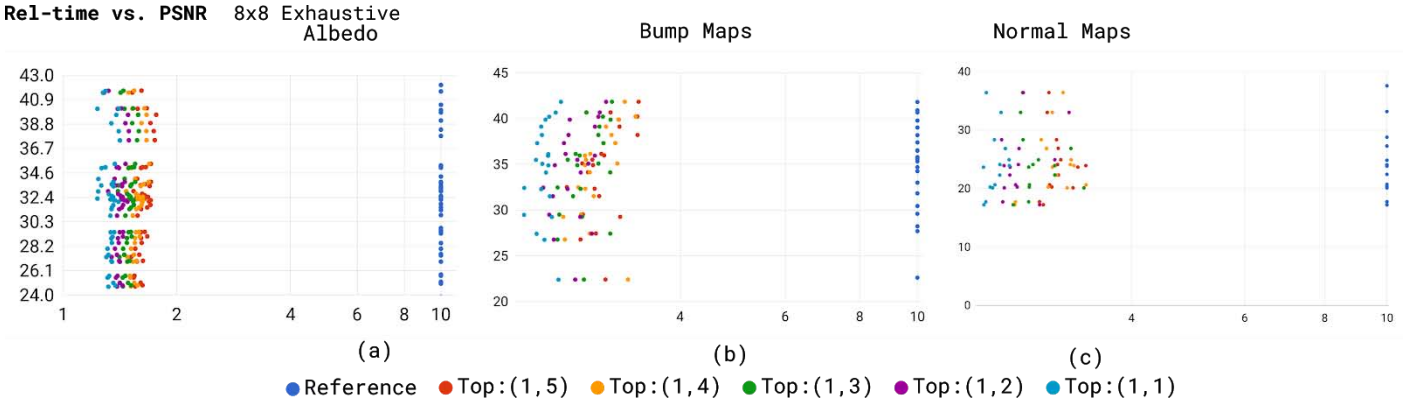


The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in through setting for block size 6x6. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.

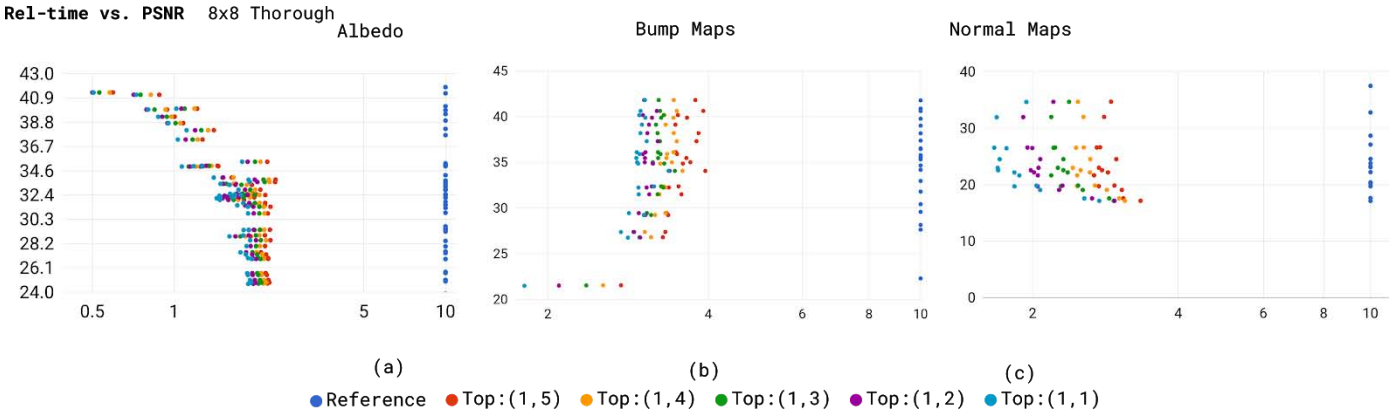


The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in fast setting for block size 6x6. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.

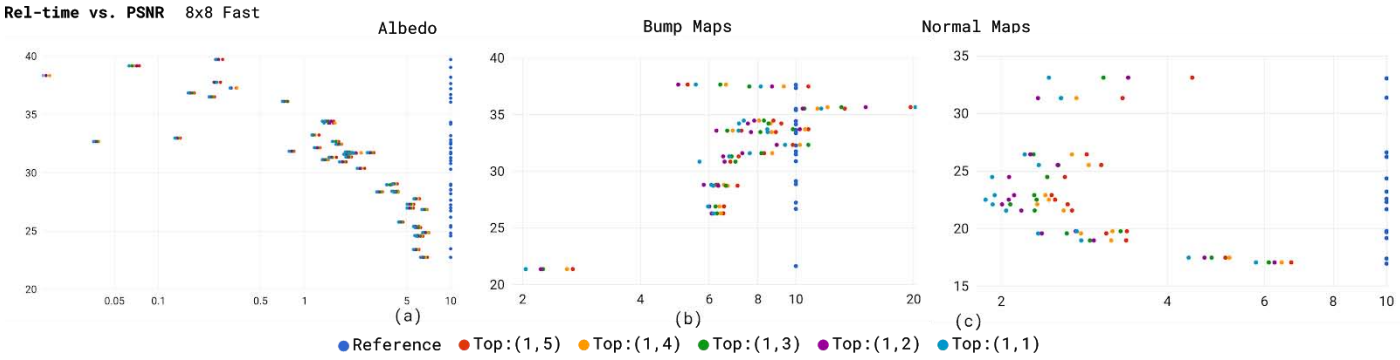
Block-size: 8x8



The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in exhaustive setting for block size 8x8. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.



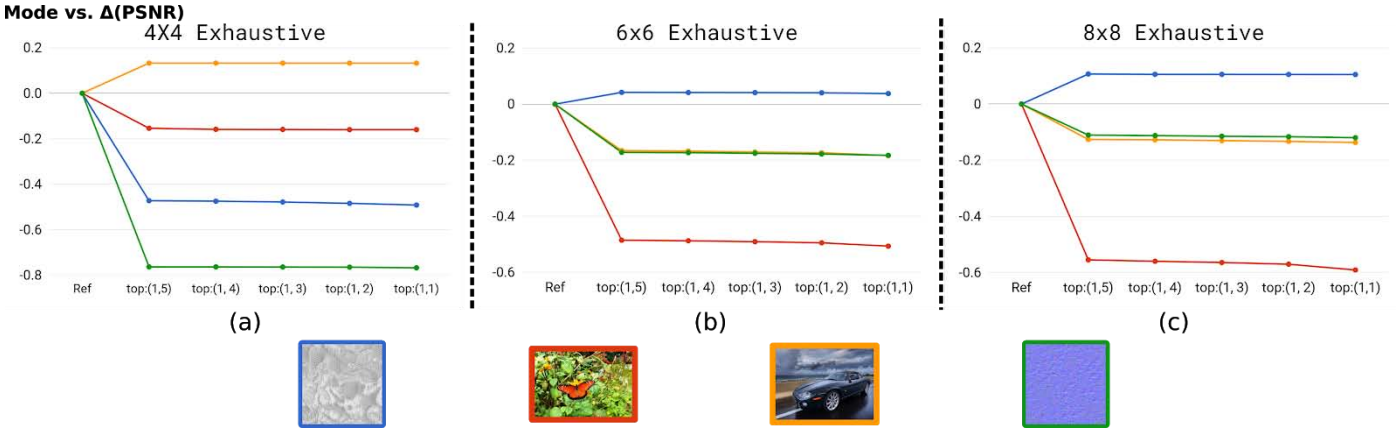
The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in thorough setting for block size 8x8. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.



The encoding time vs. quality comparison between TexNN and ref-encoder on different texture modalities in fast setting for block size 8x8. For each comparison, we fix the setting to one of the ref-encoder settings and vary TexNN settings from top:(1, 1) to top:(1, 5). The encoding speeds are measured on an Intel(R) Xeon(R) CPU, 2.4GHz using a single thread.

2.2 Change in Quality (PSNR) w.r.t ref-encoder

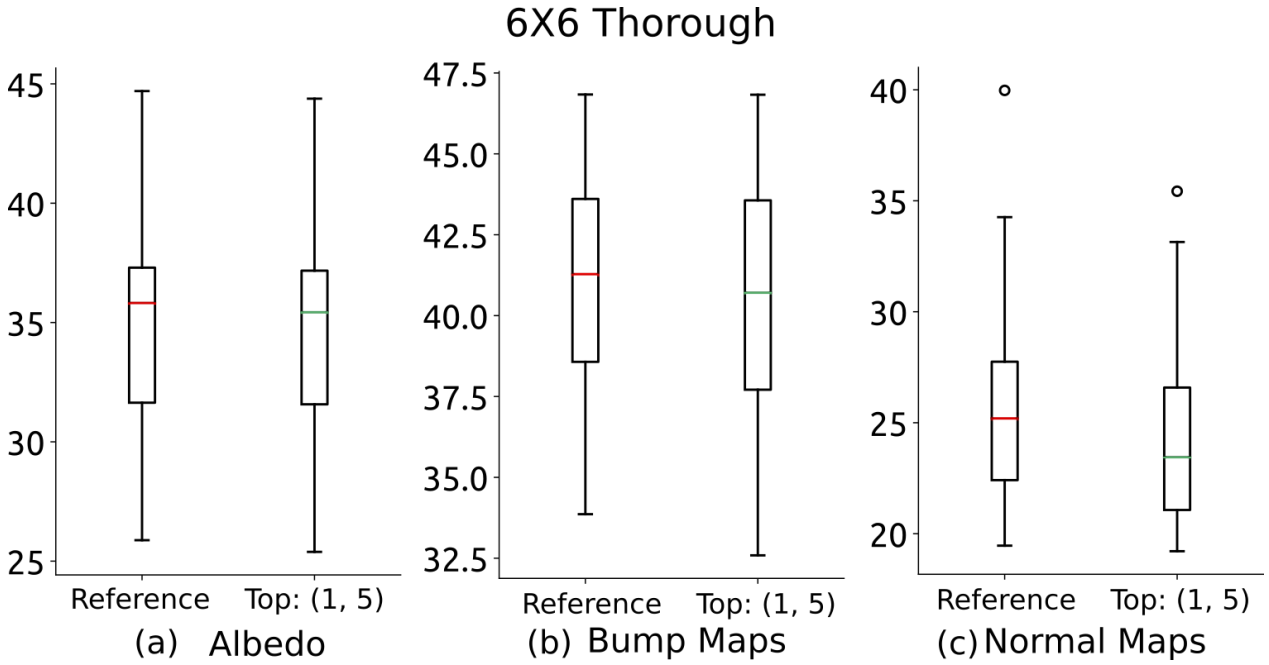
We plot the change in the PSNR with different encoding settings to highlight the change in quality. This plot is similar to Fig. 10 with mode set to exhaustive for three block sizes 4X4, 6X6, and 8X8. In some cases we notice that TexNN gives better quality than ref-encoder while the encoding speed is still around 7x faster.



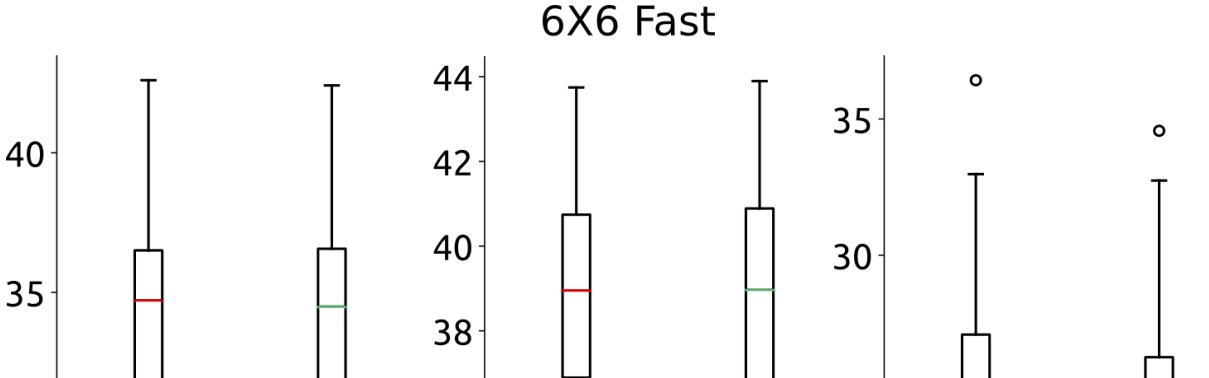
We plot the change in PSNR for different settings of TexNN from ref-encoder for a few textures from the test dataset. We show the changes for exhaustive and fast settings for 4X4, 6X6, and 8X8. Part (a) shows the plot for 4X4 block size. Part (b) plots the change for 6X6 and, we notice a little increase in quality using TexNN in the fast setting. We also notice a little increase in quality

for 8X8 in the fast setting, as seen in part (c). We observe this small increase in quality across few images for different block sizes in some encoding settings.

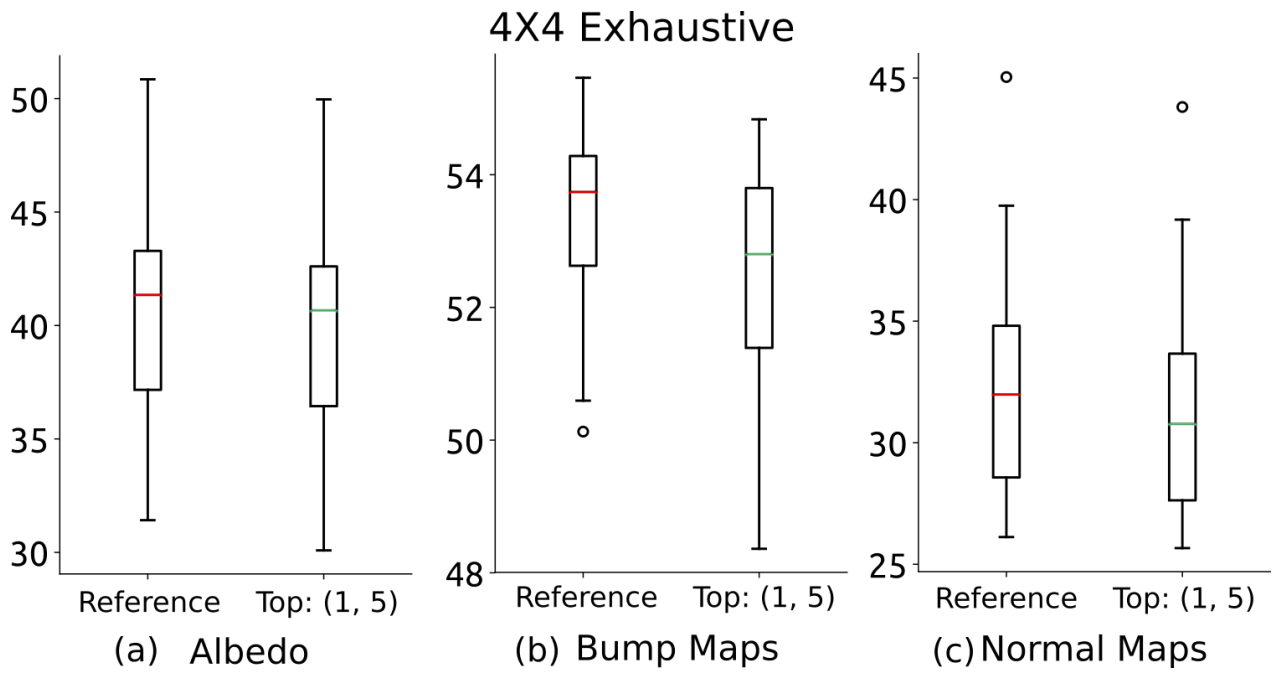
2.3 Quantitative PSNR comparison between TexNN and reference encoder



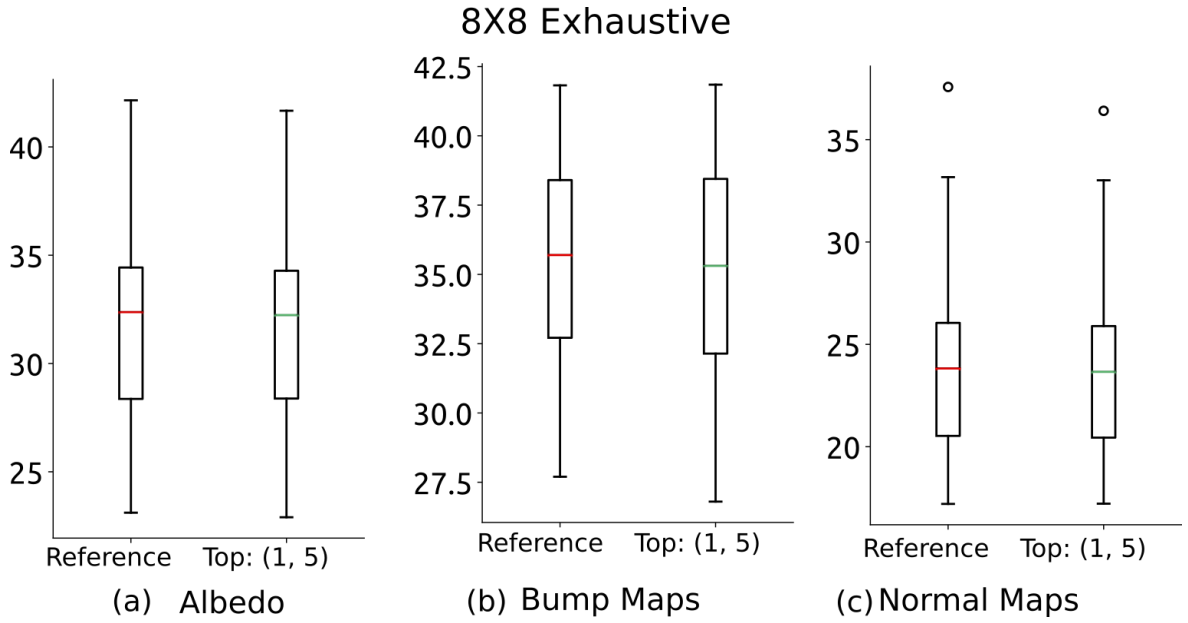
We compare the quantitative difference in the compression quality between reference encoder and TexNN using box and whiskers plot for block size of 6x6 in the through encoding-setting. It can be seen that the PSNR values for the test set are in the similar range for both TexNN and reference-encoder.



We compare the quantitative difference in the compression quality between reference encoder and TexNN using box and whiskers plot for block size of 6x6 in the fast encoding-setting. It can be seen that the PSNR values for the test set are in the similar range for both TexNN and reference-encoder.



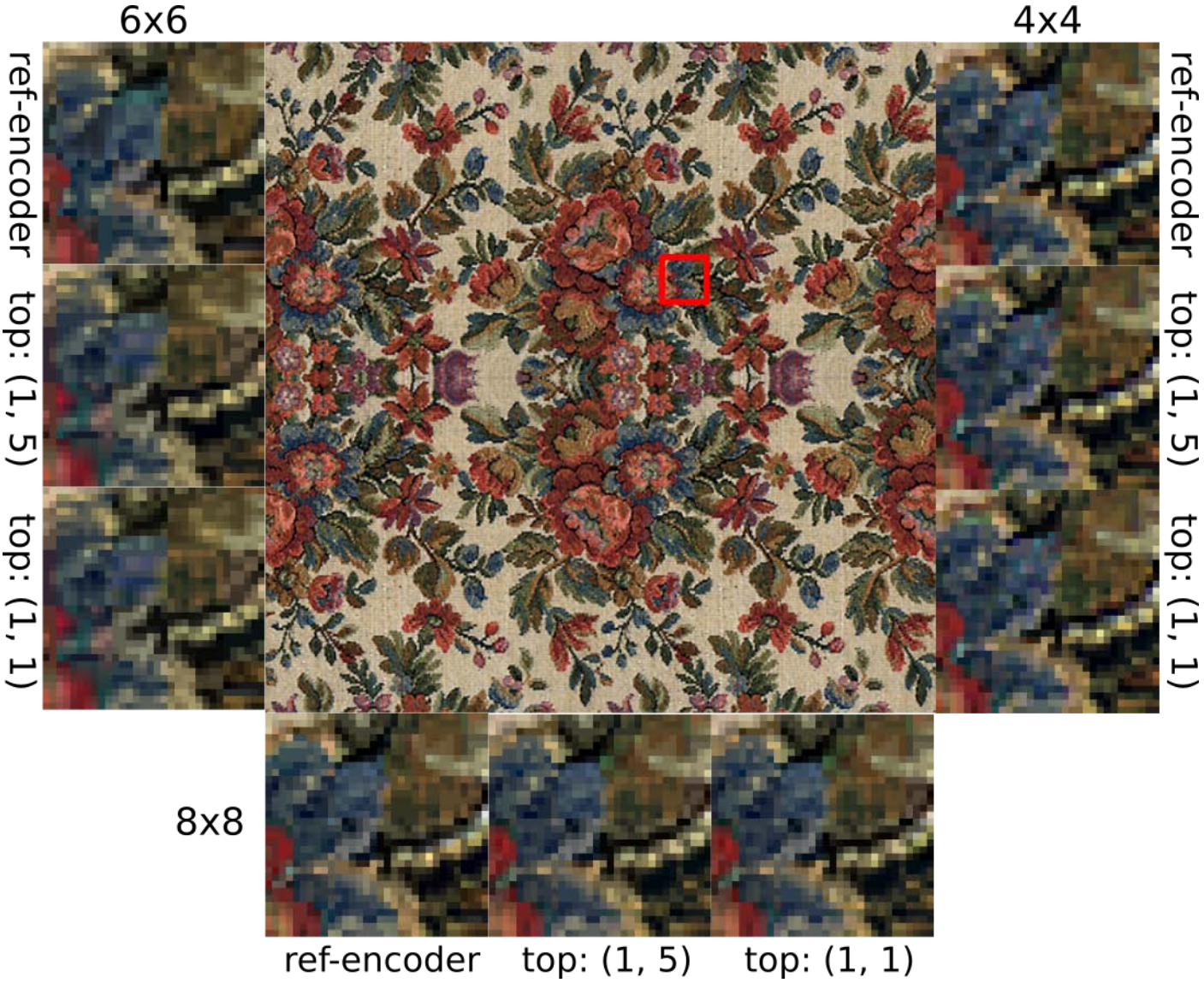
We compare the quantitative difference in the compression quality between reference encoder and TexNN using box and whiskers plot for block size of 4x4 in the exhaustive encoding-setting. It can be seen that the PSNR values for the test set are in the similar range for both TexNN and reference-encoder.



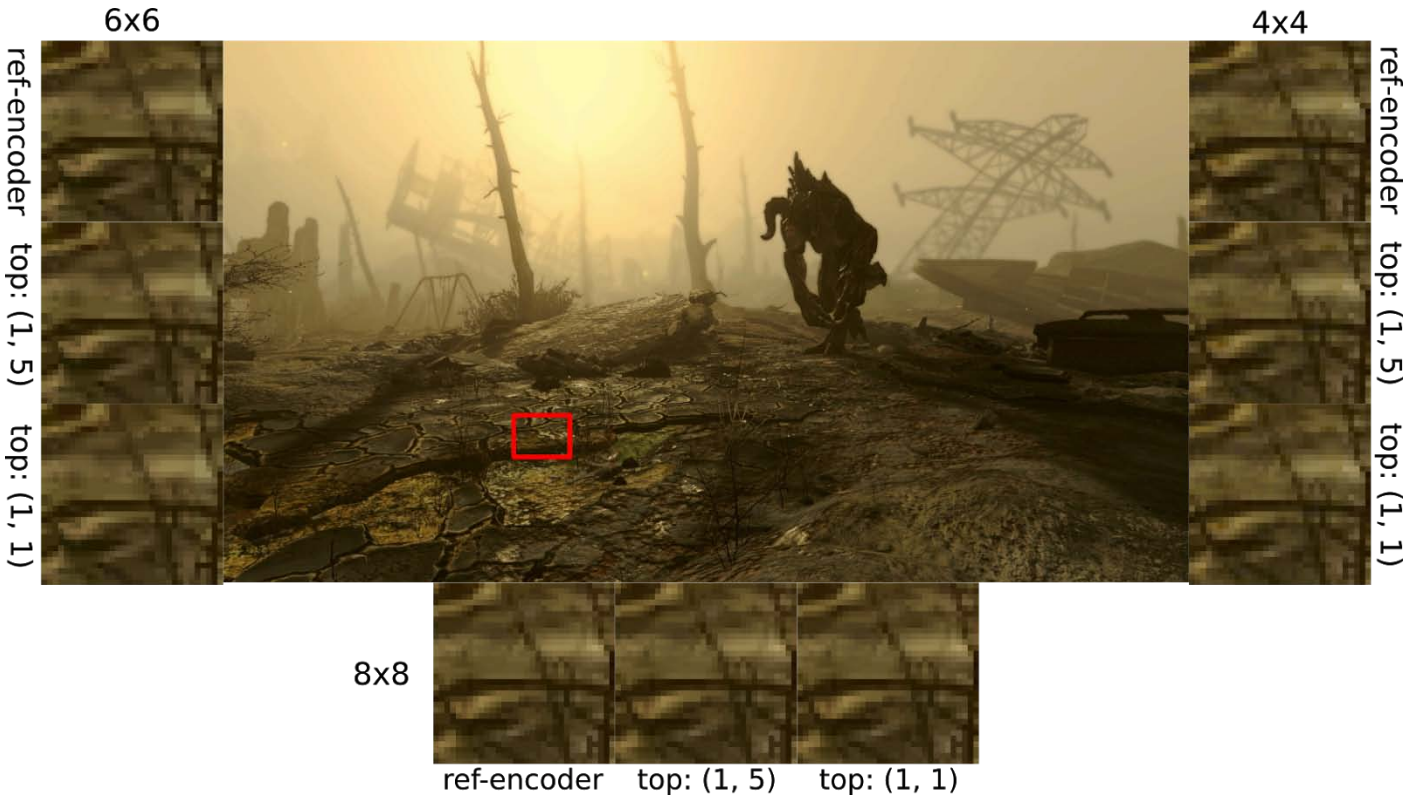
We compare the quantitative difference in the compression quality between reference encoder and TexNN using box and whiskers plot for block size of 8x8 in the exhaustive encoding-setting. It can be seen that the PSNR values for the test set are in the similar range for both TexNN and reference-encoder.

3.Visual comparison

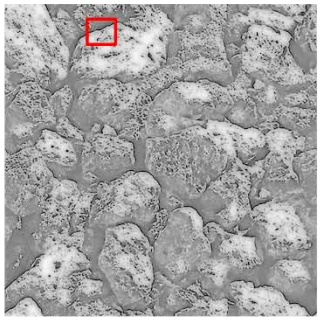
3.1 Visual Quality Comparison



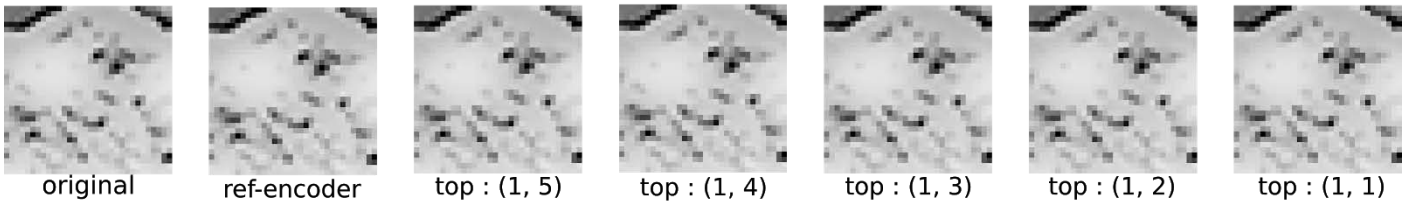
The image is taken from Pixar-128 data set mentioned in the paper. We present zoomed-in comparisons of a small 32x32 region of the image for several block sizes here. For each block-size we compare the zoomed-in versions from the reference-encoder and TexNN in top: (1, 5) and top: (1, 1). We gain a speed up of 7X with littles loss in visual quality as shown.



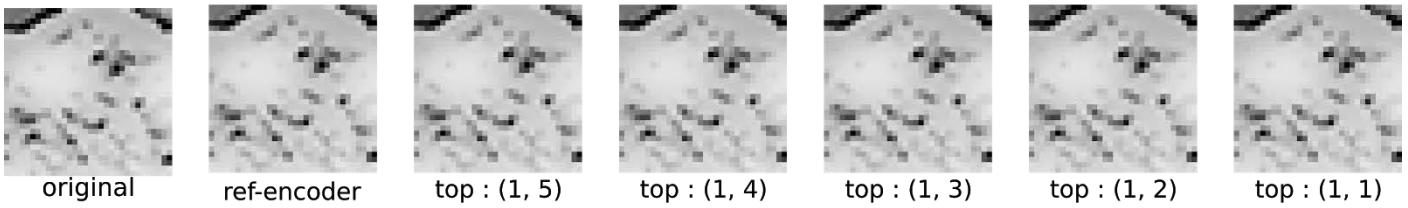
The image is taken from Fallout high resolution data set mentioned in the paper. We present zoomed-in comparisons of a small 32x32 region of the image for several block sizes here. For each block-size we compare the zoomed-in versions from the reference-encoder and TexNN in top: (1, 5) and top: (1, 1). We gain a speed up of 7X with littles loss in visual quality as shown.



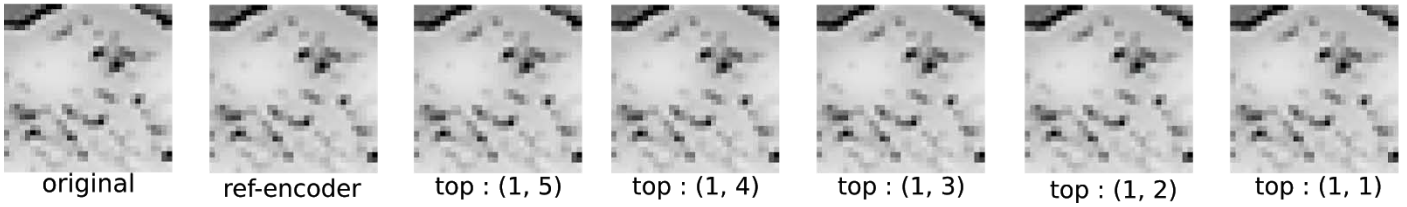
Exhaustive



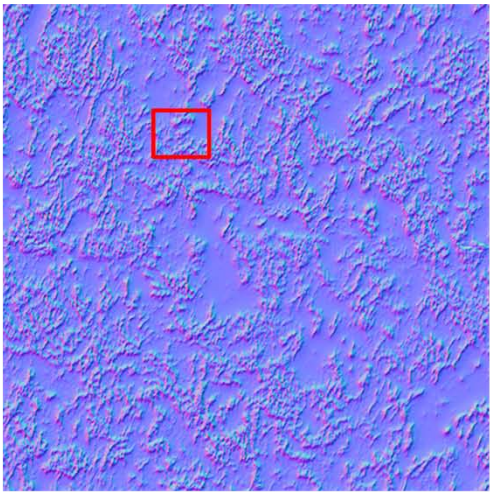
Thorough



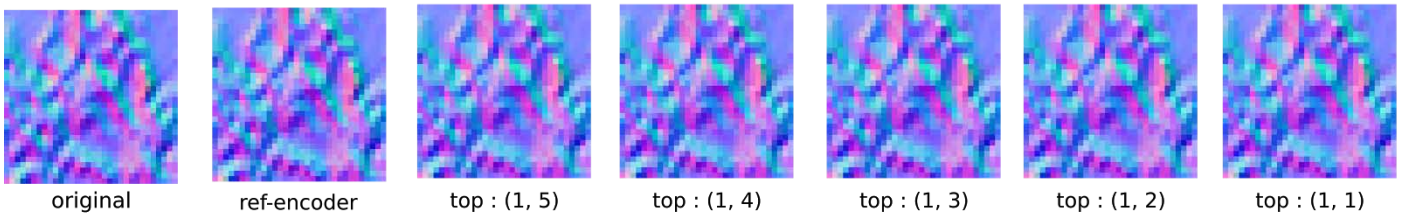
Fast



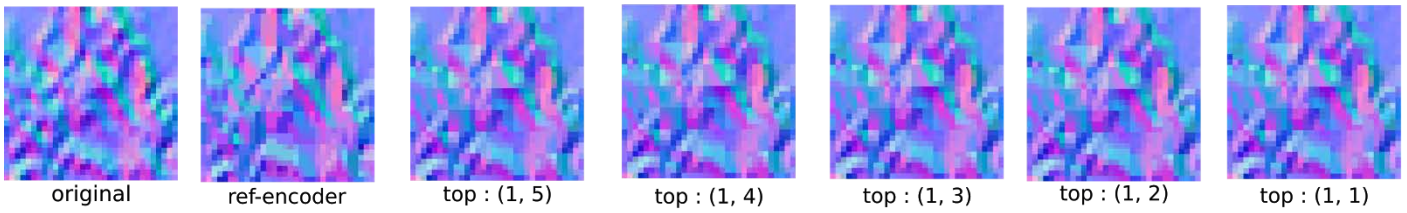
The image is taken from Pixar bumpmap dataset mentioned in the paper. We present zoomed-in comparisons of a small 32x32 region of the image for 4X4 block size here. we compare the zoomed-in versions from the reference-encoder and TexNN in top: (1, 5) to top: (1, 1). We do not notice any visual difference between compressed texture using ref-encoder and TexNN in different settings Exhaustive, Thorough, and Fast.



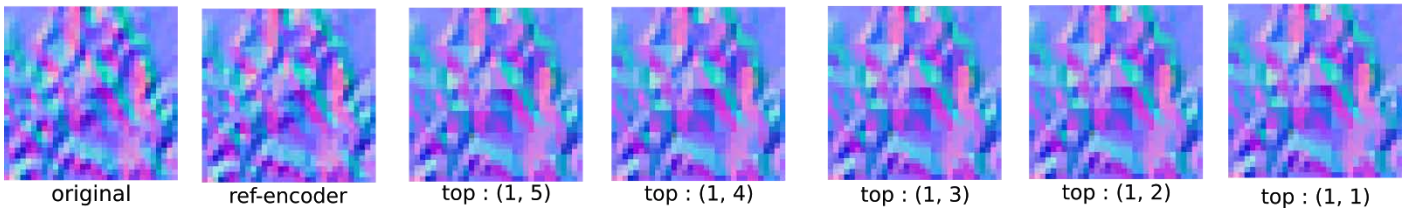
Exhaustive



Thorough



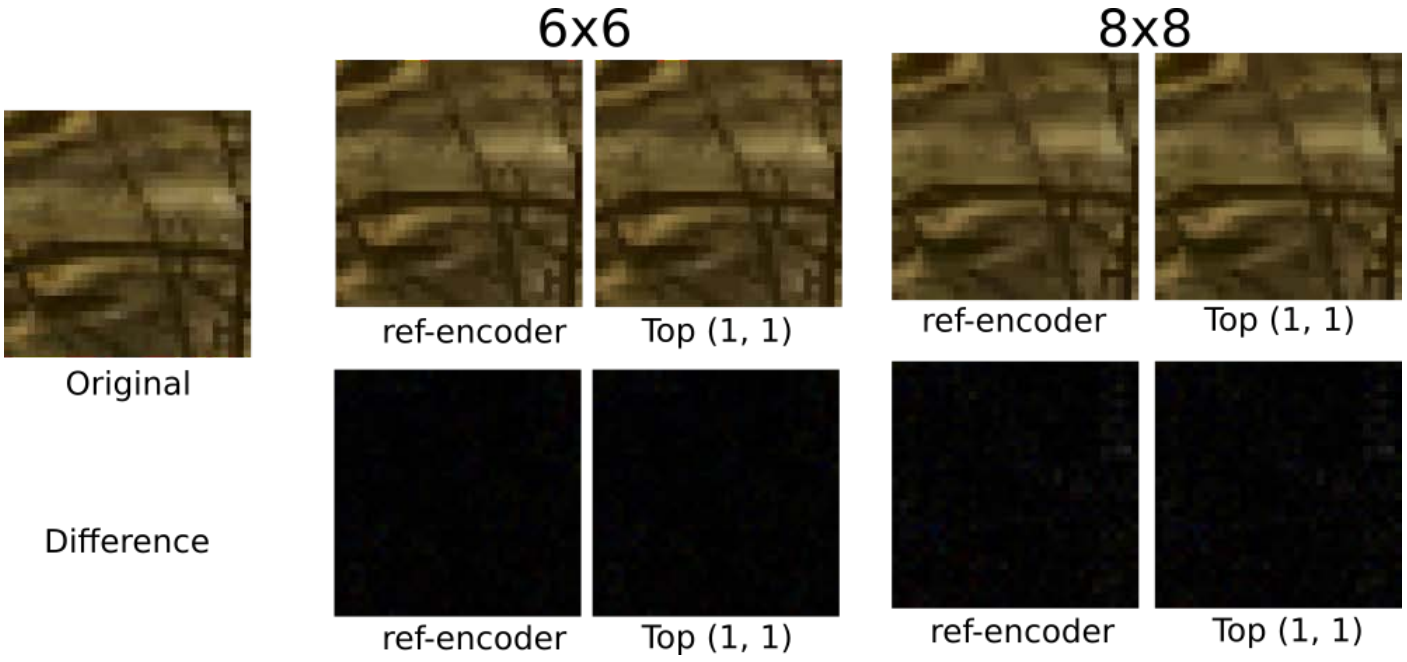
Fast



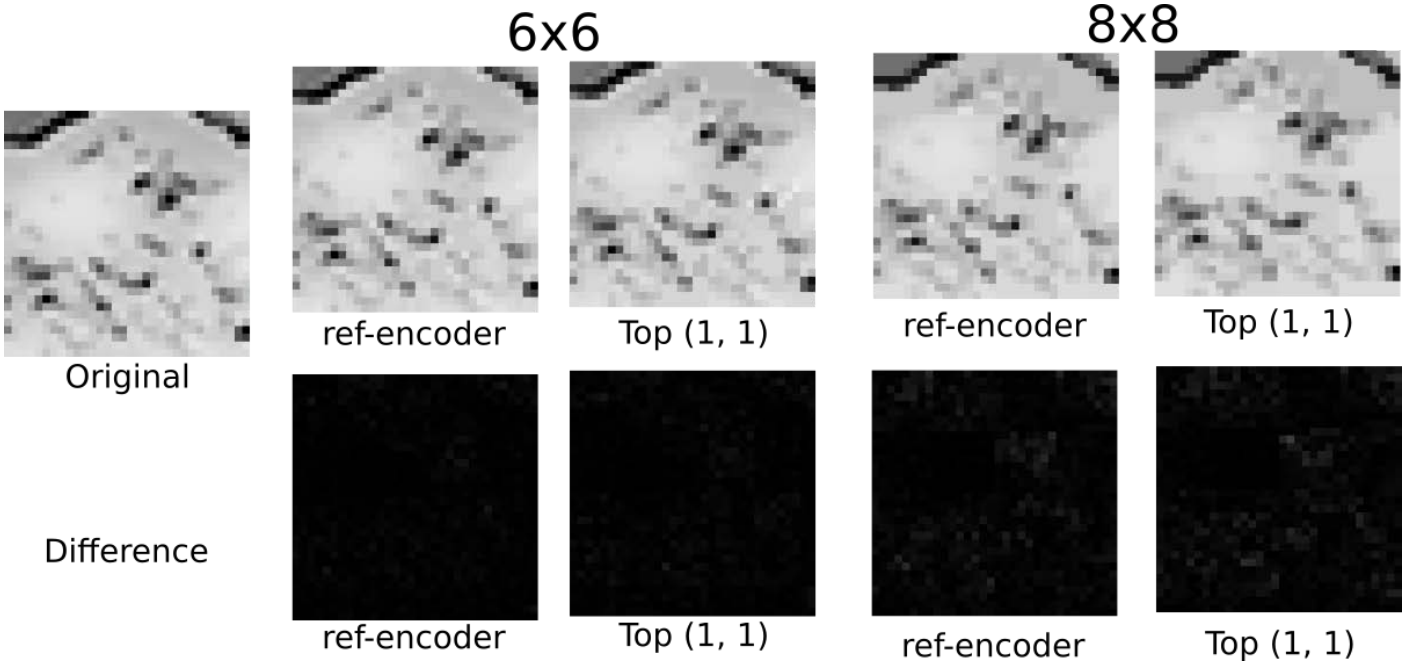
The image is taken from Pixar bumpmap dataset mentioned in the paper. We present more zoomed-in comparisons of a small 32x32 region of the image for 4X4 block size here. For each block-size we compare the zoomed-in versions from the reference-encoder and TexNN in top: (1, 5) and top: (1, 1). We do not notice any visual difference between compressed texture using ref-encoder and TexNN in different settings Exhaustive, Thorough, and Fast.

3.2 Visual Difference Comparison

In this we perform a difference between a small zoomed in regions between original image and the decompressed images. The visual difference comparison gives a more insight into the performance of TexNN.

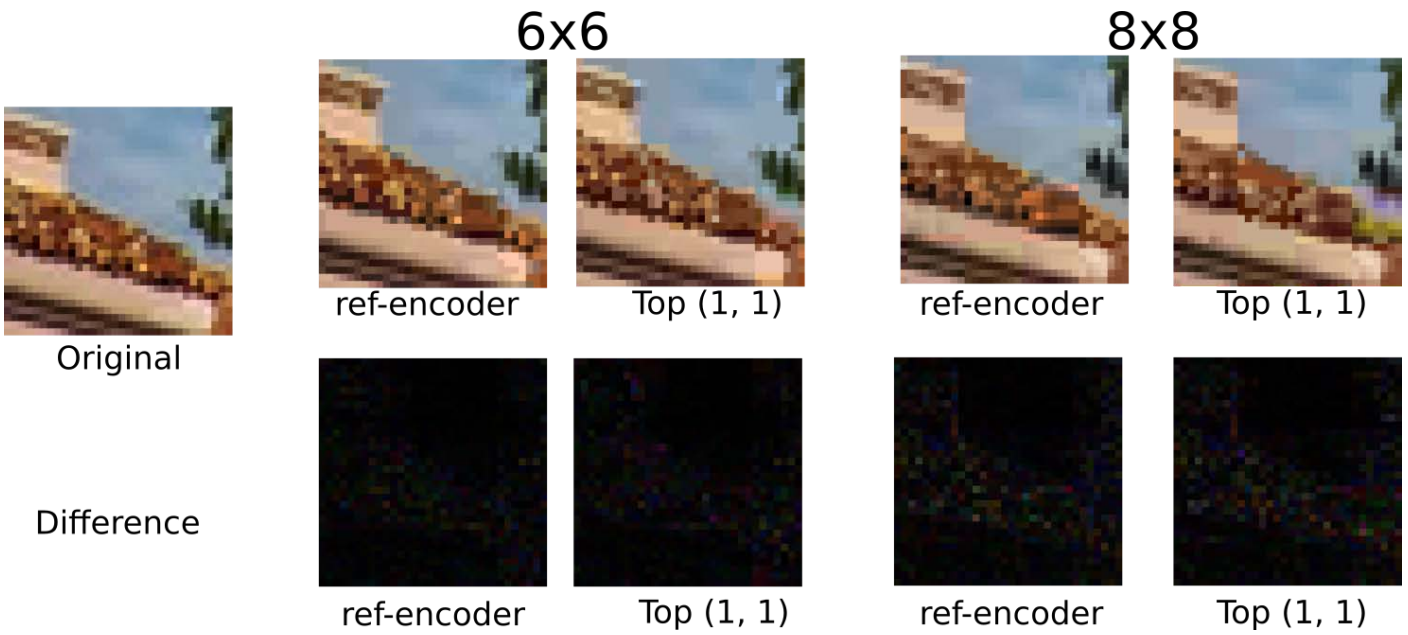


To the the left we have the scaled up version of a small 32x32 region from the original image. In the top row we show the same region from the decompressed image compressed using reference encoder and TexNN in Top (1, 1) mode. In the bottom row we show the difference image between the decompressed image and the original image. We don't see a stark change in the difference images which show that our method TexNN provides good visual quality.



To the the left we have the scaled up version of a small 32x32 region from the original image. In the top row we show the same region from the decompressed image compressed using reference encoder and TexNN in Top (1, 1) mode. In the bottom row we show the difference image between the decompressed image and the original image. We don't see a stark change in the difference images which show that our

method TexNN provides good visual quality. We present the results for two different block sizes where there is noticeable difference.



To the the left we have the scaled up version of a small 32x32 region from the original image. In the top row we show the same region from the decompressed image compressed using reference encoder and TexNN in Top (1, 1) mode. In the bottom row we show the difference image between the decompressed image and the original image. We don’t see a stark change in the difference images which show that our method TexNN provides good visual quality. We present the results for two different block sizes where there is noticeable difference.